## Experiment 5

## Average Fuel Efficiency and Percentage Improvement

### Aim

To calculate the average fuel efficiency of different car models and determine the percentage improvement between two car models using NumPy arrays.

### Procedure

1. Import the NumPy library.
2. Create a NumPy array named fuel_efficiency containing the fuel efficiency values of different car models in miles per gallon (mpg).
3. Use np.mean(fuel_efficiency) to calculate the average fuel efficiency.
4. Select the fuel efficiency values of the two car models for comparison.
5. Calculate the percentage improvement using the formula:

$$\text{Percentage Improvement} = \frac{\text{New} - \text{Old}}{\text{Old}} \times 100$$

6. Store the calculated values in separate variables.
7. Display the average fuel efficiency and percentage improvement.

### Code

```
import numpy as np

fuel_efficiency = np.array([25, 30, 35, 40, 45])

average = np.mean(fuel_efficiency)

old_model = fuel_efficiency[0]

new_model = fuel_efficiency[4]

improvement = (

   (new_model - old_model) / old_model

) * 100

print("Average fuel efficiency:", average, "mpg")

print("Percentage improvement:", improvement, "%")
```

### Result

Thus, the average fuel efficiency of the car models and the percentage improvement between the two selected models were calculated successfully using NumPy arrays and arithmetic operations.

**Output**

```
========== RESTART: D:/fundamentails of data science/experiment 5.py ==========
Average fuel efficiency: 35.0 mpg
Percentage improvement: 80.0 %
```